

ELISABETH LEUNG

(SHE/HER)

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EDUCATION

- 2024 - Present **University of California, Riverside, Riverside, CA**
Ph.D Evolution, Ecology, and Organismal Biology
- Expected graduation in Spring 2029
- 2020 - 2023 **Northeastern University, Boston MA**
B.S. Data Science and Ecology & Evolutionary Biology
- Summa Cum Laude with College of Science Honors in the Discipline
- Relevant Coursework: Urban Ecology, Biostatistics, Wildlife Biology, Ecology, Genetics and Molecular Biology, Information Presentation & Visualization, Introduction to Evolution, Machine Learning 1 & 2, Geographic Information Systems, Database Design, Foundations of Ecological & Evolutionary Genomics, Foundations of Data Science

GRANTS AND FELLOWSHIPS

Summary: 2 awards (TOTAL: \$168,100)

- 2025 **Graduate Research Fellowship, National Science Foundation (\$159,000)**
- 2024 **Donor Fund, Friends of the Island Fox (\$5,100)**
- 2022 **Marine Science Center Undergraduate Research Internship (\$4,000)**

HONORS AND AWARDS

Summary: 8 awards (TOTAL: \$64,0,00)

- 2024 **Dean's Distinguished Award, University of California Riverside (\$53,000)**
- 2024 **Marine and Environmental Science Overall Excellence Award**
- 2024 **Marine and Environmental Science Academic Excellence Award, graduated with a GPA >3.9**
- 2022 **Northeastern University Khoury College Conference Travel Support (\$500)**
- 2022 **Marine Science Center Undergraduate Research Internship (\$4000)**
- 2020 - 2023 **Dean's List Award, seven semesters at Northeastern University**
- 2020 **Woodland Park Zoo Thrive Emerging Conservationist Award (\$5,000)**
- 2020 **The Rotary Club of Bainbridge Island Community Service Recognition Scholarship (\$3,000)**
- 2020 **Bainbridge Island Sportsmen's Club Scholarship (\$2,500)**

RESEARCH EXPERIENCE

- 2024 - Present **Graduate Research Assistant, University of California, Riverside**
- Analyzed South African tiger samples by running quality control and mapping them to the reference genome
- Compiled a database of all relevant tiger samples on NCBI
- Assembled the base genome for the island fox and generated simulated founder-relatedness
- 2023 **Explore Bioinformatics, Genetics, and Genomics (BiGG) Data REEU, University of Tennessee Knoxville, Knoxville, TN**
- Conducted population genomics analysis on Atlantic White-cedar sequencing data using R and Linux
- Completed specialized training in microbiome and population genomic theories, scientific communication and leadership development, and rhizosphere DNA extractions

- 2022 - 2023 **Honors in the Discipline Research Project, Northeastern Marine Science Center, Nahant, MA**
- Conducted trials for noninvasive genotyping of juvenile Eastern Oysters
 - Tried different DNA extraction methods for different types of genetic material
 - Quantified the genetic material amount and quality with Qubit and gel electrophoresis
 - Analyzed optimal treatment and size using survival analysis in R
- 2022 **Research Intern in Marine Conservation Genomics, Lotterhos Lab, Northeastern Marine Science Center (full-time co-op), Nahant, MA**
- Conducted phenotype analysis and tissue biopsies on >900 Eastern oysters from 35 sites across the range of Texas to Nova Scotia
 - Collected oysters in four states
 - Trained in aquaculture and oyster breeding techniques
 - Extracted DNA from > 100 samples
- 2019 **Living Northwest Conservation Assistant, Woodland Park Zoo, Seattle, WA**
- Assembled and installed camera traps and scent lures across Seattle
 - Sorted 200+ camera trap pictures to track urban carnivore movements
 - Organized a research database of relevant papers
 - Released 70+ endangered butterfly larvae and 30+ endangered turtles

RESEARCH SKILLS

Laboratory Skills: DNA Extraction; Qubit; Gel electrophoresis; Osteocrust sampling; Marine invertebrate tissue biopsy; Tissue preservation for molecular analysis; PCR; RNA Extraction; Liquid Nitrogen; Autoclave;

Computer Skills: R (advanced); python (advanced); GIS (advanced); Linux; HTML; Javascript; Microsoft Office Suite; SQL; Appsheet;

Field Experience: Marsh oyster collection; Boat sample collection; Field gear organization;

Live Specimens: Aquaculture; Oyster Breeding and Experimental crosses: Sea tables and flow-through tanks; Flipping oyster aquaculture bags; Oyster relaxing protocols

TEACHING EXPERIENCE

University of California Riverside Teaching Assistant:

2025 **BIOL 105: Evolution.** Led weekly discussions on papers. Administered and graded weekly quizzes and exams. Created study guides and practice exams.

Northeastern University Teaching Assistant:

2023 **EEMB 2303: Ecology.** Teach ecological concepts and run wet lab experiments weekly. Graded weekly labs, homework assignments, and papers.

2022; 2023 **ENVR2501: Biostatistics.** Presented statistical and technical material weekly to 20 students regarding the R programming language. Graded weekly lab reports, quizzes, and projects.

PUBLICATIONS & PRESENTATIONS

- 2026 **Leung E, Lotterhos K.** A comparison of non-lethal genotyping methods for juvenile Eastern Oysters. In review at *Ecology & Evolution*.
- 2025 **Leung E, Mooney J, Armstrong A.** “Evolutionary dynamics of small population recovery in the island fox” *Evolution* 2025. Athens, GA, United States (Poster)
- 2023 **Leung E*, Degreenia O*, Dekovich A, Smith Z, Hale C, Gleason J, Staton M, Granger J, and Hadziabdic D.** “Bolstering the bog: assessing the population genomic diversity of the imperiled Atlantic white-cedar (*Chamaecyparis thyoides*)” Summer Research Scholars Symposium, Knoxville, TN, United States (Poster)
- 2023 **Leung E, Eppley M, Lotterhos K.** “Synopsis of Non-Invasive Genotyping Methods with Associated Mortality for Juvenile Eastern Oysters” RISE, Boston, Massachusetts. (Poster)
- 2022 **Leung E.** “Non-invasive Genotyping of the Eastern oyster” Marine and Environmental Science Undergraduate Research Symposium Fall 2022, Northeastern University, Nahant, Massachusetts. (Oral Presentation)
- 2022 **Leung E, Eppley M, Lotterhos K.** “Synopsis of Non-Invasive Genotyping Methods with Associated

Mortality for Juvenile Eastern Oysters” WREN Scholar Summit, Boston, Massachusetts. (Poster)

2022 **Leung E**, Eppley M, Lotterhos K. “Synopsis of Non-Invasive Genotyping Methods with Associated Mortality for Juvenile Eastern Oysters” SSMPG 2022, Aussios, France. (Poster)

WORKSHOPS

2024 Island Fox Necropsy Workshop, Santa Barbara Zoo, Santa Barbara, California

2024 Ancient DNA Analysis Workshop, American Museum of Natural History, New York City, New York.

2022 Software and Statistical Methods for Population Genomics (SSMPG), Aussios, France.

2022 Evolution in Changing Seas Research Coordinated Network (RCN) Training and Integration Workshop, Shoals Marine Lab, Appledore Island, Maine.

VOLUNTEER, SERVICE, AND LEADERSHIP

2025-2026 **Graduate Student Association Outreach Coordinator**, University of California Riverside, Riverside, CA

- Coordinate with external organizations that need volunteers or science outreach
- Organize and create scientific outreach for different outreach events

2024-2025 **Department Newsletter Coordinator**, University of California Riverside, Riverside, CA

- Communicate with the department to collect research, outreach, and teaching accomplishments
- Compile all the results in the bi-yearly departmental newsletter

2020-2023 **Peer Educator**, Northeastern University Alliance of Civically Engaged Students, Boston MA

- Researched, created, and presented social justice presentations monthly to first-years
- Guided 8 first-year students transitioning into college through a year-long commitment at a non-profit organization
- Facilitated bi-weekly group lessons and reflections on social justice issues affecting Boston

2022-2023 **Peer Mentor**, Marine and Environmental Sciences Peer Mentoring, Northeastern University, Boston, MA

- Led biweekly topical meetings for four students
- Answered questions about coop and classes

2022-2023 **Member**, Alpha Phi Omega (National Service Fraternity), Northeastern University, Boston, MA

- Served 20+ hours of community service every semester at various organizations in the Greater Boston area
- Selected for recruitment and finance committees

2023 **Climate Justice Fellow**, Northeastern University Climate Justice & Sustainability Hub, Boston, MA

- Completed specialized training in structural oppression, racial justice, and climate & environmental inequality trainings
- Reflected and discussed concepts and issues with other fellows
- Led community conversations with organizations to work towards just solutions

2017-2021 **Zoo Ambassador**, Woodland Park Zoo, Seattle, WA

- Presented educational programming about conservation, climate change, pollution, and animal behavior to zoo guests
- Led a team of teens that developed materials designed to teach guests conservation-related crafts and messaging
- Organized a zoo-wide climate action event with a team of four students
- Completed 500+ community service hours over three years

OUTREACH

2024 **P-22 Day**, Los Angeles, CA

- Assisted Friends of the Island Fox in educating about the island fox conservation status
- Performed mock health checks on island foxes for interested visitors

- 2022 **Showcase of Opportunities for Undergraduate Research (SOURCE)**, Northeastern University, Boston, MA
- Presented Lotterhos Lab's current and past research to prospective undergraduate students
 - Answered all questions regarding the lab, co-op, and future research endeavors
- 2022 **North Andover Field Trip**, Northeastern University Marine Science Center, Nahant, MA
- Assisted Marine Science Center outreach department with 40 AP Biology and Marine Biology students to deliver an intertidal lab
 - Answered questions from students about my research, classes, and Northeastern
- 2022 **Beach Sisters Tour**, Northeastern University Marine Science Center, Nahant, MA
- Presented my research and career development opportunities to middle school and high school students
 - Answered questions about co-op and undergraduate experiences

WORK EXPERIENCE

- 2024 **Dog Walker**, Happy Dogs Boston
- Walked and managed up to 20 dogs daily in pairs while maintaining a high level of care and attention to detail
 - Created and sent behavioral reports to owners after walks
- 2023 **Launch Coordinator**, Northeastern University Alliance of Civically Engaged Students, Boston, MA
- Planned a week-long orientation to Boston through community service and civic engagement events
 - Coordinated with local organizations for civic learning events and catering
 - Managed social media and training booklets

MENTORSHIP

- 2026 **Sam Anderson**; undergraduate at University of California Riverside
- 2025 **Olive Levesque**; undergraduate at Mt. San Antonio College (UCRNRS Summer Intern)
- 2022-2023 **Michael Budner, Ava DiMauro, Noah Lugani, Veronica Rajotte**; undergraduates at Northeastern University
- 2022 **Owen Hong**; undergraduate at Northeastern University